Comparative Time Series Forecasting of Oil Prices

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Github Link: https:/github.com/thulluri-ramya/Comparative-Time-Series-Forecasting-of-Oil-Prices

1. Introduction:

Oil prices have a big effect on the world economy since they change the cost of transportation, energy production, and all economic activity. Because of this, governments, investors, and businesses that make strategic and financial decisions need to be able to reliably predict fluctuations in oil prices. But it's hard to guess what oil prices will be because they are affected by many unknown things, such as geopolitical events, changes in supply and demand, and changes in the worldwide market.

Time series forecasting techniques are extensively employed to examine previous data and predict future outcomes. People often utilize traditional statistical approaches like Autoregressive Moving Average (ARMA) and Autoregressive Integrated Moving Average (ARIMA) models to look at time series data [2].

Recently, more and more people have been using machine learning and deep learning to solve problems with making predictions. When it comes to sequential data, Long Short-Term Memory (LSTM) networks have done better than other types of networks [1]. LSTM models can learn complex patterns and long-term correlations from time series data [1], which makes them great for modelling economic and financial variables.

This project aims to analyse and predict oil price movements through statistical and deep learning methodologies. We create and apply an ARIMA model and an LSTM neural network to look at oil price data. We utilize typical forecasting error measures to measure how well both models work, and we use both models to make predictions for the next 24 months. After that, the results are compared to see what each modelling method is good at and what it can't do.

2. Data Description and Pre-Processing:

The dataset used in this analysis contains historical observations of oil prices measured in US dollars over time. Each observation corresponds to a specific date and the associated price of oil on that date. Because the data is structured chronologically, it is suitable for time series analysis and forecasting.

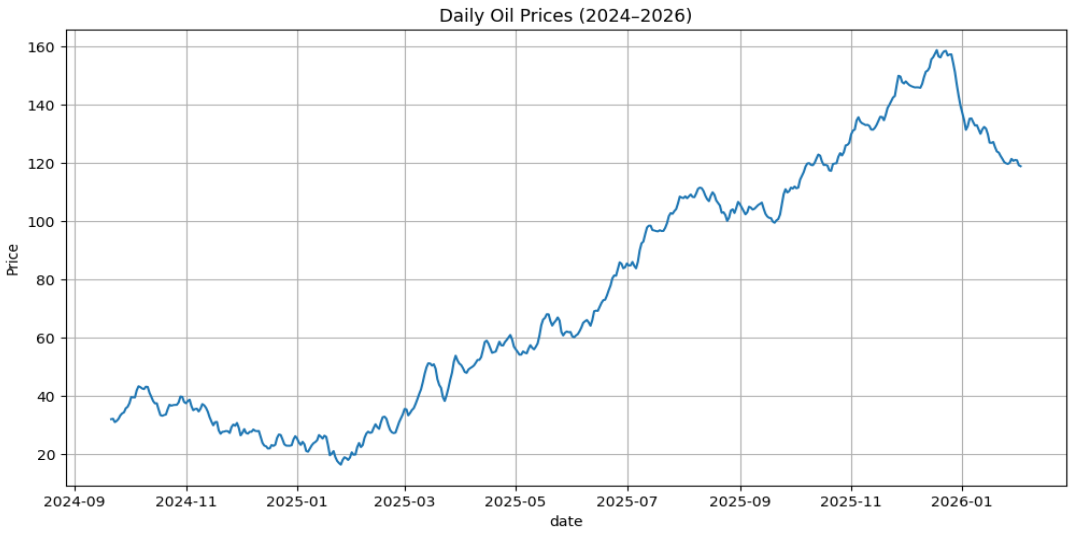

Figure 1: Historical oil price time series used for modelling.

The dataset was prepared for predicting models by going through a number of preparation processes. The date column was changed to datetime format and utilized as the dataset's index. This step is very important because it makes it possible to process the data correctly as a time series. We then made a visual study of the oil price series by graphing the data over time. This helped to find general patterns, such as trends, changes, and times of high volatility. We also used basic descriptive statistics to help us understand what the dataset was like. These numbers indicate the average, standard deviation, lowest, and highest prices for oil. These summary metrics help us understand how the data is spread out and how much it changes.

Table 1: Summary Statistics

count 500.000000	mean 75.873980
std 42.912398	min 16.480000
25% 35.372500	50% 64.290000
75% 111.572500	max 158.780000

An extra preprocessing step was necessary for the LSTM model. Neural network models typically exhibit enhanced performance when input values are normalized to a comparable range. Consequently, the oil price values were normalized utilizing a MinMaxScaler, which converted the data to a range between 0 and 1. This scaling enhances the model's stability and efficiency during the training phase.

3. Stationarity Analysis:

Stationarity is essential for numerous traditional time series models, such as ARIMA [2]. A stationary time series keeps its statistical properties the same over time, which means that the mean and variance stay the same. The model's results may not be reliable if the time series is not stationary.

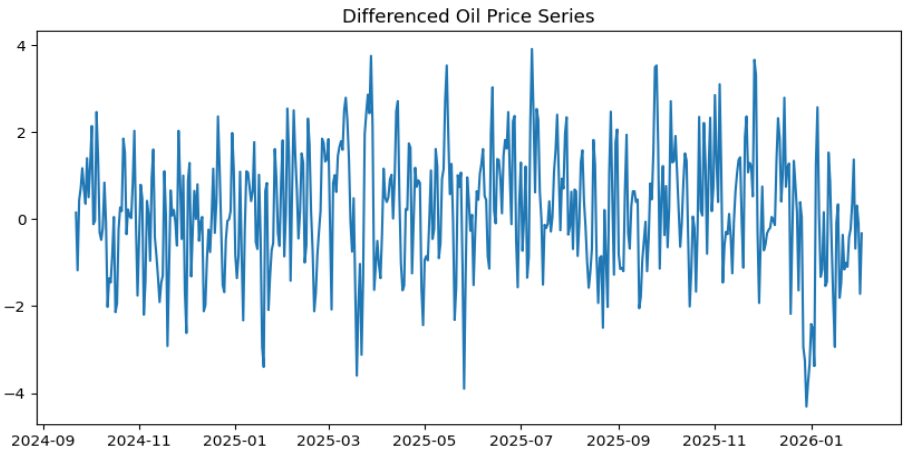

Figure 2: Differenced oil price series used to achieve stationarity.

We used the Augmented Dickey-Fuller (ADF) test to see if the oil price data were stable. The null hypothesis for the ADF test posits that the time series is non-stationary owing to the existence of a unit root. If the final p-value is more than 0.05, the data is termed non-stationary, and the null hypothesis cannot be rejected.

The ADF test results indicated that the initial oil price series lacked stationarity. This is a common result for financial time series because they often show patterns and levels that change over time. To fix this problem, the data was first-order differenced. Differencing works by taking the difference between the current observation and an earlier one. This helps to get rid of the trend and make the series more stable.

The data's stationarity was checked again after differencing was used. The transformed series exhibited enhanced stability, making it appropriate for modelling with the ARIMA methodology. We also made graphs of the Autocorrelation Function (ACF) and the Partial Auto Correlation Function (PACF) to assist us pick the best model parameters.

4. ARIMA Model:

The ARIMA model is a well-known statistical method for looking at and predicting time series data [2]. There are three parts to the model: autoregressive (AR), differencing (I), and moving average (MA). The autoregressive part shows how an observation is related to its prior values, while the moving average part shows how an observation is related to past forecast errors. When necessary, the differencing component is used to make the series stationary.

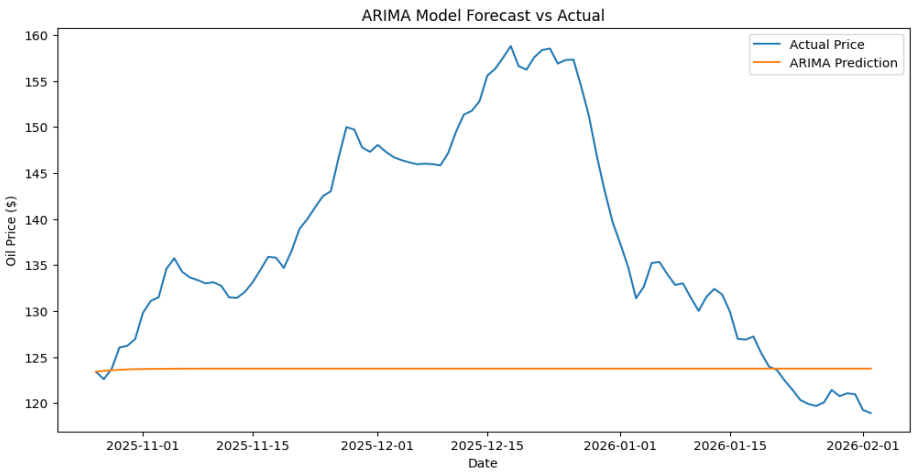

Figure 3: ARIMA predictions compared with actual oil prices.

This study utilized the ARIMA model on the oil price dataset subsequent to verifying the stationarity of the differenced series. We looked at the ACF and PACF plots to find the right values for the moving average and autoregressive parameters. Based on this analysis, the model's parameters were chosen to work well together.

A grid search method was used to find the optimal ARIMA configuration by looking at different combinations of parameters. The letters p, d, and q stand for the orders of autoregressive, differencing, and moving average. The parameter space was examined with p values ranging from 0 to 8, d values from 0 to 2, and q values from 0 to 8. For every feasible combination, the Akaike Information Criterion (AIC) was worked out. The model with the lowest AIC value was chosen as the best fit because it strikes a better balance between fit and complexity.

We looked at how close the model's predictions were to the actual oil prices to see how accurate it was.

Table 2: ARIMA Evaluation Metrics

MAE	14.572486526233384
RMSE	18.047123733676408

We used two metrics to see how well the model worked: mean absolute error (MAE) and root mean squared error (RMSE). These numbers show the average difference between the expected values and the actual data. Lower values for these variables mean that the predictions are more accurate.

The trained ARIMA model was employed to generate forecasts for the ensuing 24 months. Confidence intervals were included to show how unpredictable these future projections are.

Residual Analysis:

After we fitted the ARIMA model, we conducted some tests on the residuals to determine whether it had done a decent job of illustrating how the data was set up. The residuals should seem like white noise, with a normal distribution and no clear autocorrelation. We generated an autocorrelation function (ACF) plot of the residuals to see if there were any correlations that might still be in the model errors. The model accurately depicted the primary temporal structure of the series, evidenced by the absence of significant spikes in the residual autocorrelation function (ACF). A histogram was used to see if the errors were almost normally distributed by checking the distribution of the residuals.

5. LSTM Model:

An LSTM neural network was also employed to predict oil prices, along with the statistical ARIMA model. LSTM networks are a type of recurrent neural network that are made to cope with data that comes in a series [1]. LSTM models may remember information from earlier time steps, which lets them find long-term relationships in time series data. This is different from regular neural networks.

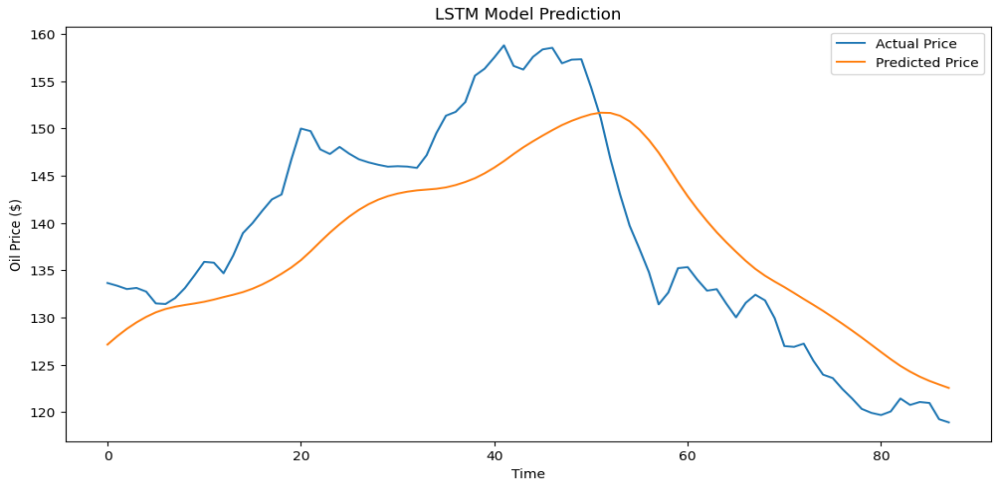

Figure 4: LSTM model predictions compared with actual values.

We employed a MinMaxScaler to normalize the oil price data within the range of 0 to 1 before to training the LSTM model. Subsequently, input sequences for the model were generated via a sliding window technique. This method utilized a fixed quantity of prior observations, referred to as the window size, to predict the subsequent value in the series. This work employed an LSTM architecture with two LSTM layers followed by a dense output layer. We employed the Adam optimization algorithm for model training, with the loss function configured as mean squared error. The algorithm analysed patterns and correlations in historical oil price data throughout its training phase.

Subsequent to its training, the model was employed to generate predictions for the test dataset. The anticipated values were reverted to their original size and juxtaposed with actual oil prices. The LSTM model was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), similar to the ARIMA model.

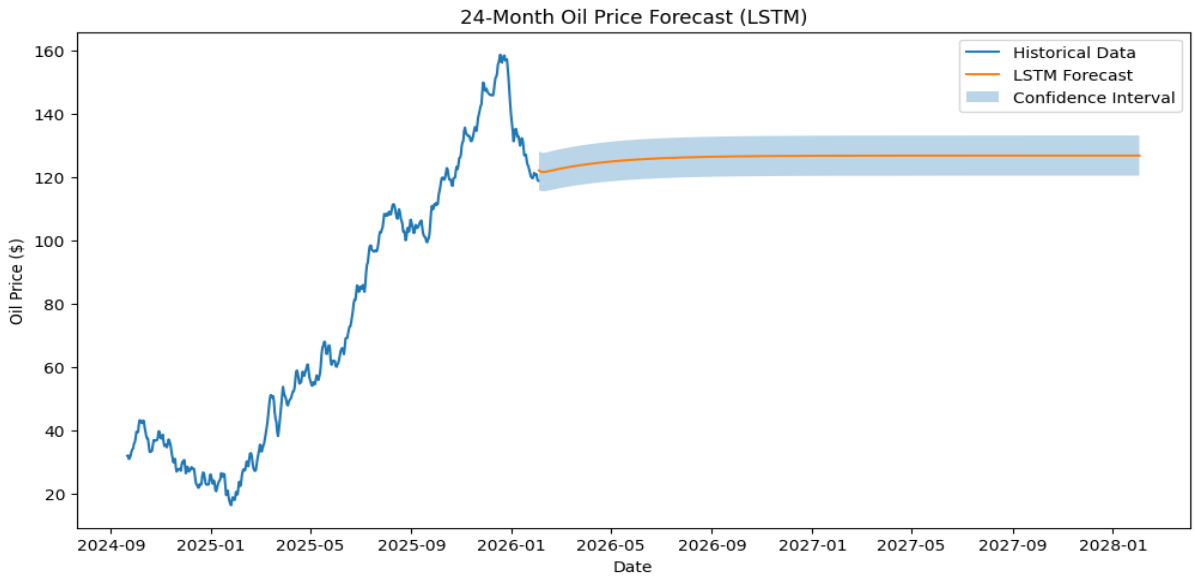

Figure 5: 24-month oil price forecast generated using the LSTM model.

Finally, the trained LSTM model was used to forecast oil prices for the next 24 months, allowing for a comparison between the statistical and deep learning approaches.

6. Results and Model Comparison:

After implementing both the ARIMA and LSTM models, their forecasting performance was evaluated using standard error metrics. The models were tested using the test portion of the dataset, and their predictions were compared with the actual observed oil prices.

Table 3: comparison table from both model Results.

Model	**MAE**	**RMSE**
ARIMA	14.57	18.05
LSTM	6.41	7.23

We used two evaluation criteria to check how accurate the forecasts were: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). MAE calculates the average absolute distinction between expected and actual values, while RMSE gives more weight to larger errors and shows how accurate the forecast is overall.

The results show that the LSTM model had less errors than the ARIMA model. This means that the LSTM model did a better job of capturing the patterns in the oil price time series. The LSTM model's higher performance might be explained by its ability to detect complex and nonlinear relationships in sequential data.

Along with a numerical evaluation, the predicting ability was also looked at visually by comparing the anticipated values to the actual observations. The ARIMA model was able to see the general trend in the oil price series, but it didn't always do a good job of following rapid changes. The LSTM model, on the other hand, made predictions that were more in line with the changes in the data.

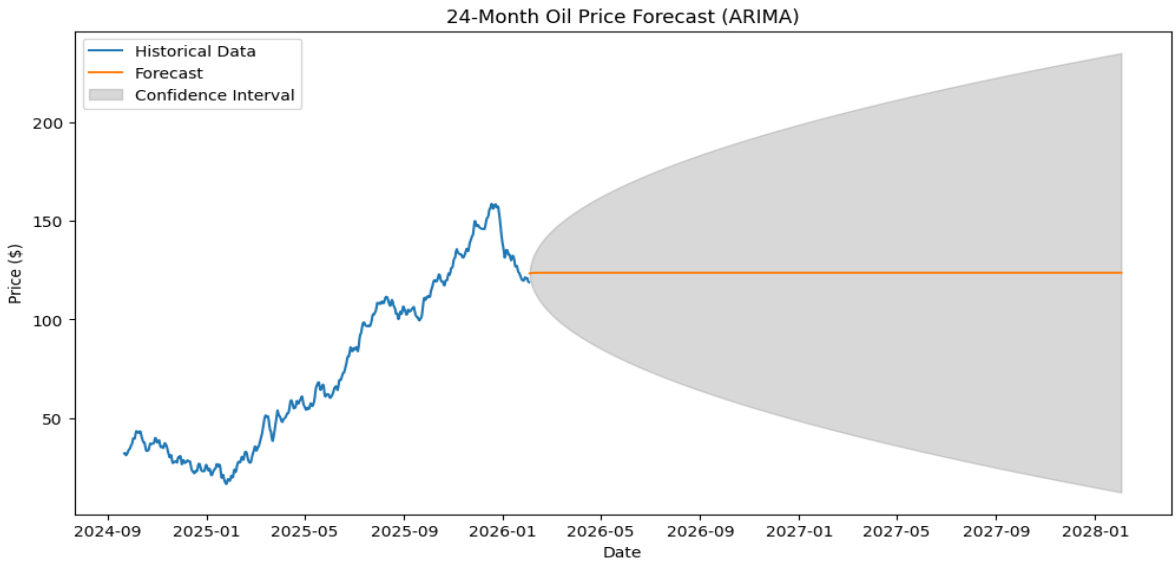

Figure 6: 24-month oil price forecast generated using the ARIMA model.

Both models were employed to generate forecasts for the next 24 months. The ARIMA forecast included confidence intervals, illustrating the uncertainty associated with long-term predictions. The breadth of the confidence intervals widened over time, signifying the growing uncertainty in predicting future results.

When we compared the predicted values to the real oil prices in the testing dataset, we saw that the LSTM model did a better job of following the changes in the series than the ARIMA model. The ARIMA model successfully discerned the primary trend of the time series but struggled to react swiftly to sudden variations in oil prices. In contrast, the LSTM model exhibited enhanced adaptability to short-term variations, resulting in diminished forecasting errors.

7. Discussion and Critical Analysis:

Although both the ARIMA and LSTM models were able to generate reasonable forecasts, their performance differed due to the nature of their modelling approaches.

The ARIMA model is based on linear relationships within the time series data. While it performs well when patterns are stable and the data is stationary, it may struggle to capture complex nonlinear behaviour. Oil prices are influenced by many unpredictable factors [3], such as geopolitical events, global demand changes, and economic policies. Because ARIMA relies primarily on historical patterns, it may not fully capture these dynamic influences.

On the other hand, the LSTM model is capable of learning nonlinear relationships and long-term dependencies within the data. This allows the model to adapt more effectively to the fluctuations commonly observed in financial time series. The improved accuracy of the LSTM model in this study suggests that deep learning approaches may be more suitable for modelling complex economic variables such as oil prices.

Another important observation from the forecasts is the increase in uncertainty as the forecast horizon becomes longer. This is reflected in the widening confidence intervals in the ARIMA forecasts. Such behaviour is expected in time series forecasting because prediction errors accumulate over time.

Despite the advantages of the models us ed in this study, several limitations should be acknowledged. The analysis was based solely on historical oil price data without incorporating additional variables that may influence price movements. Factors such as global economic conditions, supply disruptions, and geopolitical conflicts can significantly impact oil markets. Including such external variables could potentially improve forecasting performance.

8. Future Work and Improvements:

There are several ways in which this analysis could be improved in future work. One possible improvement would be to incorporate additional explanatory variables that influence oil prices, such as global oil demand, inflation rates, or economic growth indicators. Including these variables could help capture broader economic dynamics that affect oil price movements.

Another potential improvement would be to experiment with more advanced deep learning architectures. Models such as Gated Recurrent Units (GRU) or transformer-based time series models have shown promising results in recent forecasting studies and may provide improved predictive performance.

Furthermore, ensemble approaches that combine predictions from multiple models could be explored. By integrating the strengths of different modelling techniques, ensemble methods [4] often produce more robust and accurate forecasts.

9. Conclusion:

This research examined both conventional and deep learning approaches for predicting oil prices. The two models utilized were the ARIMA model and the Long Short-Term Memory (LSTM) neural network.

The results showed that both models could accurately show the overall trend of the oil price time series. The LSTM model, on the other hand, did a better job of predicting when it came to error metrics and its ability to follow data changes. This shows that deep learning methods could be useful for working with nonlinear and complicated time series data. The results suggest that combining modern machine learning techniques with traditional statistical methods can provide valuable insights for forecasting time series data. No model can fully predict how oil prices will change, but these methods can help make predictions more accurate and help people make better choices.

Even though the LSTM model did a better job of predicting accuracy in this study, statistical models like ARIMA are still important because they are easier to understand and cost less to run.

10. References:

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